CO2 PROBLEM SOLVING ASSIGNMENT

Abstract:
This report develops a practical transmitter for an eight-channel dense wavelength-division multiplexed (DWDM) link at 1550 nm. Each wavelength carries 25 Gb/s NRZ over six amplified 80 km fiber spans. Instead of choosing a source based only on its output power, the design checks emission wavelength, quantum efficiency, modulation speed, temperature drift, nonlinear effects, soliton conditions and integration options. The calculations show that a wavelength-locked DFB laser with external modulation is a sensible starting point, but several requirements still need laboratory and standards verification before deployment.

Interpret the communication requirement :
Eight 25 Gb/s channels provide 200 Gb/s aggregate raw line rate. Their nominal spacing is 100 GHz, or about 0.8 nm near 1550 nm. Six 80 km spans give a 480 km optical path. Long-distance transmission needs narrow-linewidth, single-frequency lasers to prevent adjacent-channel interference. The transmitter also needs adequate optical power, low chirp, controlled wavelength and reliable high-speed modulation. Amplifiers restore power after each span but do not undo accumulated distortion or amplified spontaneous-emission noise.

- Bandgap sets the approximate emission photon energy; internal quantum efficiency describes radiative versus total recombination.
- Threshold current and slope efficiency determine useful output above lasing threshold; linewidth and side-mode suppression ratio (SMSR) affect channel isolation.
- Laser chirp, chromatic dispersion, relative-intensity noise, extinction ratio and temperature drift influence the received eye opening.
- Fiber loss, effective length, nonlinear coefficient, dispersion, channel spacing and launch power determine nonlinear interactions.
- ITU-T G.957 concerns optical interfaces for SDH systems; IEEE 802.3bm and 802.3bs concern specified Ethernet PHY interfaces. Neither automatically certifies this custom 8×25 Gb/s 480 km DWDM line.

- Eye safety must be assessed against the applicable ANSI Z136 exposure and laser-classification framework, with wavelength, exposure duration, aperture and fault conditions defined.

Problem requirement statement: Build eight stable 1550 nm optical channels, each carrying 25 Gb/s NRZ, with sufficient optical power, at least 17.5 GHz nominal electrical modulation bandwidth, controlled chirp and wavelength drift, acceptable nonlinear penalties, safe accessible emissions and a scalable transmitter architecture.

Emission wavelength and internal efficiency:

Photon emission occurs when electrons and holes recombine radiatively; non-radiative recombination wastes carrier energy as heat. Given Eg = 0.80 eV: λ (μm) $\approx$ 1.24/Eg = 1.24/0.80 = 1.55 μm = 1550 nm.

Given radiative lifetime τr = 10 ns and non-radiative lifetime τnr = 40 ns: ηint = $(1/\tau r)/[(1/\tau r)+(1/\tau nr)] = \tau nr/(\tau r+\tau nr)$ = 40/(10+40) = 0.80 = 80%. This is the radiative recombination fraction in the simplified lifetime model, not the complete wall-plug efficiency.

Source comparison:

Source	Spectrum / speed	Power, threshold and coupling	Assessment
LED	Broad spectrum; limited direct modulation speed	No lasing threshold; weak coupling to single-mode fiber	Unsuitable for 100 GHz long-haul DWDM
FP laser	Multiple longitudinal modes; fast variants exist	Lasing threshold; good guided coupling	Mode competition makes dense channels difficult
DBR laser	Narrow spectrum; tunable variants	Good power and fiber coupling; temperature/current tuning	Strong alternative when tunability is important
DFB laser	Single longitudinal mode; narrow linewidth; fast	Stable wavelength with feedback; strong single-mode coupling	Preferred source for fixed-grid channels

VCSEL	Compact and efficient; fast variants	Low threshold; convenient arrays; 1550 nm options need qualification	Interesting for short reach or qualified 1550 nm arrays
QD laser	Potential low chirp and temperature sensitivity	Performance varies greatly with platform and maturity	Promising alternative after device-level validation

Laser oscillation begins when round-trip optical gain equals round-trip cavity loss. For a DWDM source, high SMSR suppresses unwanted longitudinal modes; low linewidth limits spectral overlap and phase-noise penalties. Exact SMSR and linewidth targets must be set from the modulation format and receiver design, not invented from the given data.

Design Decision 1: I recommend a 1550 nm DFB laser for each fixed DWDM channel because it supports narrow-linewidth, single-mode emission, good fiber coupling and stable wavelength locking. A tunable DBR laser is a credible alternative when remote channel reconfiguration or wavelength inventory reduction is required.

Modulation, chirp and wavelength stability

Laser output and NRZ bandwidth

Ith = 12 mA, I = 40 mA, ηs = 0.3 mW/mA. $P = \eta s(I - Ith) = 0.3 \times (40 - 12) = 8.4$ mW. In dBm: $P(dBm) = 10 \log10(8.4/1) = 9.24$ dBm. This is the estimated laser output before modulator and coupling losses, not the 2 mW per-channel fiber launch.

Required NRZ bandwidth $B \approx 0.7Rb = 0.7 \times 25$ GHz = 17.5 GHz. Given 3 dB bandwidth = 20 GHz; margin = 20 − 17.5 = 2.5 GHz (approximately 14.3% above the required bandwidth). The simplified bandwidth test passes, although eye-mask and rise-time tests are still necessary.

Rate equations and relaxation oscillation:

A standard simplified laser model is $dN/dt = I/(qV) - N/\tau n - G(N,S)S$ and $dS/dt = \Gamma G(N,S)S + \beta sp\, N/\tau r - S/\tau p$, where N is carrier density, S is photon density, Γ is confinement factor, τp is photon lifetime and G is optical gain. The relaxation-oscillation frequency is associated with the coupled carrier–

photon response and often increases approximately with the square root of optical power above threshold. Small-signal modulation is a linearized response around bias; large-signal modulation introduces pattern effects, turn-on delay, overshoot and stronger chirp. A numeric relaxation frequency cannot be determined without gain and photon-lifetime parameters.

Direct versus external modulation:

Method	Benefit	Limitation / decision
Direct laser modulation	Low component count and potentially low cost	Carrier-driven wavelength chirp can interact with chromatic dispersion over 480 km
External Mach–Zehnder modulation	Lower transmitter chirp when properly biased; good high-speed optical quality	Additional insertion loss, driver power and bias-control complexity

Uncontrolled drift: $\Delta\lambda = (0.1\ \text{nm/°C})(40\ °\text{C}) = 4.0$ nm, equivalent to about five 0.8 nm channel spacings. To hold the drift within ±0.1 nm, $|\Delta T| \leq 0.1/0.1 = \pm 1\ °\text{C}$ relative to the set point, assuming the stated coefficient and ignoring ageing. A TEC with thermistor feedback and wavelength locker is preferred. Injection-current tuning is useful for fine trimming but changes optical power, chirp and bias; it should not replace thermal stabilization.

Design Decision 2: I recommend external Mach–Zehnder modulation and closed-loop TEC plus wavelength-locker control because 20 GHz nominal bandwidth exceeds 17.5 GHz, external modulation reduces chirp risk, and uncontrolled 4 nm drift is far too large for 0.8 nm spacing. An output-power controller must account for modulator insertion loss so the fiber launch remains at the intended 2 mW.

Nonlinearity and eye safety

Effective length and self-phase modulation

Convert attenuation to nepers per km for the power equation: $\alpha = (\ln 10/10) \times 0.2 = 0.04605\ \text{km}^{-1}$. For L = 80 km: $L_{eff} = [1 - \exp(-0.04605 \times 80)]/0.04605 = 21.17$ km. (Using 0.2 directly as km^{-1} would be incorrect because the input is in dB/km.)

With $\gamma = 1.3\ W^{-1}km^{-1}$ and $P = 2\ mW = 0.002\ W$: $\varphi NL,span = \gamma PLeff = 1.3 \times 0.002 \times 21.17 = 0.0550$ rad. Approximate six-span total $= 6 \times 0.0550 = 0.330$ rad if each span is restored to the same launch power. This simplified estimate excludes channel-dependent amplification, pulse peaks and inter-channel effects.

XPM, FWM, SBS and SRS:

XPM arises when power fluctuations in neighboring WDM channels change the refractive index seen by a channel. FWM creates new frequency components from combinations of channel frequencies and can cause crosstalk if products fall on DWDM channels. XPM and FWM increase with launch power; dispersion and channel spacing influence phase matching and temporal walk-off. Standard dispersive fiber usually reduces FWM efficiency compared with near-zero-dispersion fiber.

SBS threshold: $Aeff = 80\ \mu m^2 = 80 \times 10^{-12}\ m^2$; $Leff = 21.17\ km = 21170\ m$. $Pth,SBS \approx 21Aeff/(gBLeff) = 21(80\times10^{-12})/[(5\times10^{-11})(21170)] = 0.00159\ W \approx 1.59\ mW$.SRS threshold:$Pth,SRS \approx 16Aeff/(gRLeff) = 16(80\times10^{-12})/[(6\times10^{-14})(21170)] = 1.008\ W \approx 1008\ mW$.

At 2 mW/channel, the stated narrow-line SBS estimate is below launch power, indicating SBS needs careful checking. The total eight-channel launch is $8 \times 2 = 16\ mW = 12.04\ dBm$, well below the simple SRS threshold estimate. The SBS threshold equation is a narrow-line, effectively continuous-wave approximation; high-speed data modulation and optical linewidth broadening can substantially increase the practical SBS threshold. Total WDM power should not simply be compared with a single-tone SBS threshold.

Effect	Origin / behavior	Mitigation
SPM	A channel changes its own optical phase	Moderate launch power, dispersion-aware DSP
XPM	Neighboring channels impose phase changes	Optimize launch power, dispersion map and channel allocation
FWM	Kerr mixing generates new frequencies	Avoid low-dispersion phase matching; optimize grid and power

SBS	Narrowband backward Brillouin scattering	Linewidth broadening when compatible, monitoring and launch optimization
SRS	Broadband Raman power transfer, often shorter-to-longer wavelengths	Power tilt control, gain equalization and total-power management

Eye-safety assessment: 16 mW is the sum of guided optical powers, not an accessible emission classification. ANSI Z136 permissible exposure depends on wavelength, exposure duration, beam geometry and measurement aperture; there is no defensible universal 16 mW pass/fail limit. As an illustrative conservative engineering control, require the connector-accessible emission to meet the applicable product classification (often assessed under IEC 60825-1) with automatic shutdown on open connectors. Formal ANSI Z136 evaluation is pending; do not claim compliance from total guided power alone.

Design Decision 3: I recommend controlled per-channel launch power, narrow optical filtering and gain equalization, plus an SBS-specific linewidth/return-loss check and automatic laser shutdown. This retains useful signal power while addressing a 0.330 rad simplified six-span SPM phase and an SBS threshold estimate near the nominal 2 mW channel power.

Solitons, DSP, standards and integration

Soliton calculations

An optical soliton is possible when anomalous group-velocity dispersion and Kerr self-phase modulation balance for an appropriate pulse shape and peak power. Given $T0 = 15$ ps and $|\beta 2| = 20$ ps²/km: $LD = T0^2/|\beta 2| = 225/20 = 11.25$ km.

Fundamental soliton requires $LNL = LD$. $P0,soliton = 1/(\gamma LD) = 1/(1.3 \times 11.25) = 0.0684$ W = 68.4 mW peak.

At the stated $P0 = 2$ mW: $LNL = 1/(\gamma P0) = 1/(1.3 \times 0.002) = 384.6$ km. $N^2 = LD/LNL = 11.25/384.6 = 0.02925$; $N = 0.171$. Since N is much less than 1, the given power and pulse do not support a fundamental soliton. NRZ data

also do not naturally provide the isolated sech-shaped pulses assumed by this simple soliton model.

Digital backpropagation:

Digital backpropagation (DBP) numerically reverses the estimated dispersive and Kerr propagation of fiber, typically using split-step Fourier processing after coherent reception. It is computationally demanding and needs accurate channel and span information. Full-field compensation of all eight WDM channels is particularly difficult because it requires sufficiently broad joint-bandwidth observation. For the specified 25 Gb/s NRZ system, conventional chromatic-dispersion management and receiver equalization are the first practical options. Single-channel DBP may be evaluated only if coherent receiver architecture and measured nonlinear penalties justify it.

Standards applicability

Document	Relevant checks	Scope caution
ITU-T G.957	Optical interface application code, wavelength, launch power, spectral characteristics and receiver parameters where applicable	An SDH optical interface recommendation, not a direct specification for arbitrary 8-channel 480 km DWDM
IEEE 802.3bm / 802.3bs	Relevant 100G/200G/400G Ethernet PHY PMD and optical interface requirements where a chosen PHY is used	Eight independent 25 Gb/s DWDM wavelengths do not automatically implement a compliant Ethernet PHY
ANSI Z136 / IEC 60825-1	Accessible exposure, laser classification, labeling, interlocks and failure modes	Compliance requires wavelength-specific testing and actual accessible-emission measurements

Integration options:

Architecture	Advantages	Trade-offs

Discrete DFB + external modulator	Mature, individually replaceable, straightforward characterization	Larger footprint, more optical connectors and alignment
Monolithic InP	Laser and active optical functions on one platform; compact multi-channel design	Process complexity, thermal crosstalk and fabrication yield
Si photonics + hybrid InP laser	Scalable passive routing and modulators with integrated electronics potential	Laser attachment, packaging, coupling loss and thermal engineering

Design Decision 4: I recommend a discrete DFB-plus-external-modulator design for the first verified 8-channel prototype, with a later migration to hybrid InP/Si photonics if cost and packaging targets justify it. I do not recommend soliton transmission at 2 mW, nor full-band DBP as a default requirement. Validate dispersion compensation and nonlinear penalties experimentally before adding complex DSP.

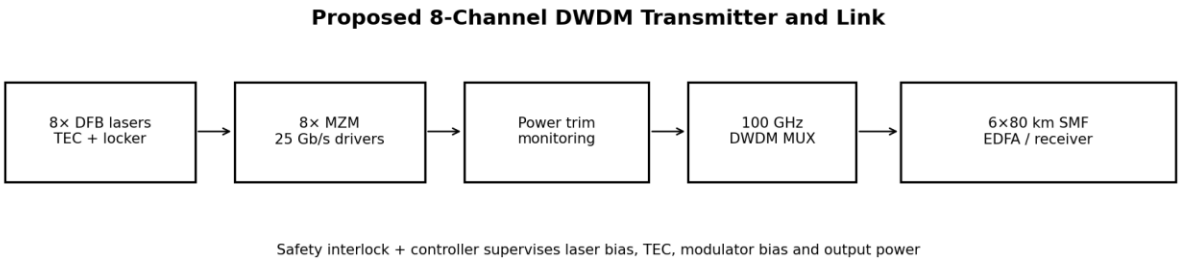
Verification:

Requirement	Target / calculated	Status
Source	1550 nm DFB; ηint = 80%	Conceptually suitable; linewidth/SMSR must be specified
Modulation and power	17.5 GHz needed; 20 GHz available; 8.4 mW raw laser output	Pass on simplified bandwidth; optical launch after modulator to be verified
Wavelength/chirp	4 nm drift without control; TEC target ±1 °C	Needs wavelength-lock and eye-diagram testing
Nonlinearity and safety	Leff 21.17 km; φ6span 0.330 rad; SBS 1.59 mW; SRS 1.008 W	SBS and accessible-emission tests outstanding
Soliton / dispersion	LD 11.25 km; N = 0.171	Soliton not feasible at given power
Integration / DSP / standards	Discrete prototype; dispersion compensation;	Conditional; not yet certified

	applicable interface and safety tests	

Final transmitter design:

Parameter	Adopted design
Channels / grid	8 channels, 100 GHz nominal spacing around 1550 nm
Data rate / distance	25 Gb/s NRZ/channel, 6 × 80 km = 480 km
Laser	DFB, nominal 1550 nm; eight wavelength-set devices
Laser operating point	Ith 12 mA; bias 40 mA; estimated pre-modulator output 8.4 mW (9.24 dBm)
Modulation	External Mach–Zehnder modulator; 20 GHz candidate 3 dB bandwidth
Wavelength control	TEC + wavelength locker; temperature error within ±1 °C by simple drift criterion
Fiber launch	2 mW per channel (3.01 dBm); 16 mW aggregate
Nonlinear controls	Power optimization, optical filtering, SBS evaluation, gain equalization
Dispersion / DSP	Dispersion compensation/equalization; no default full-band DBP
Integration	Discrete initial design; hybrid Si-photonics as scale-up option
Safety / standards	ANSI Z136 risk assessment, IEC laser classification, selected interface-standard testing



The eight wavelength-locked DFB lasers feed external modulators, optical power monitors and a 100 GHz multiplexer. The combined signal passes through six amplified fiber spans. Feedback maintains wavelength and power while a safety controller disables emission during faults. A receiver-side dispersion/equalization plan must be included in the complete link budget.

Final engineering decision:

Decision: NO for immediate field deployment; YES for controlled prototype development. The source wavelength, efficiency, power and bandwidth calculations support the proposed architecture, and the soliton result correctly rules out a simple soliton solution. However, the narrow-line SBS estimate is close to the intended launch power, temperature stability must be demonstrated, and actual eye-safety and interface compliance have not been measured. Before release, the team must complete a 480 km OSNR/BER link budget, modulator eye-mask and chirp tests, SBS/FWM/XPM characterization, wavelength-lock cycling, accessible-emission certification and reliability testing.

Conclusion:

A DFB laser with external modulation and closed-loop wavelength control is the most defensible starting design for this eight-channel DWDM link. The calculations establish 1550 nm emission, 80% simplified internal efficiency, 8.4 mW estimated laser output, 2.5 GHz nominal modulation margin, 4 nm uncontrolled thermal drift, 0.330 rad approximate accumulated SPM phase, and a soliton order of 0.171. The engineering conclusion is conditional because a workable optical design still needs verified noise, dispersion, nonlinear, safety and standards performance.

References:

- G. P. Agrawal, Fiber-Optic Communication Systems, 4th ed., Wiley, 2023.
- G. P. Agrawal, Nonlinear Fiber Optics, 6th ed., Academic Press, 2025.
- ITU-T Recommendation G.957, Optical interfaces for equipments and systems relating to the synchronous digital hierarchy.
- IEEE Std 802.3bm and IEEE Std 802.3bs, Ethernet optical PHY amendments; consult the exact selected PHY clauses.
- ANSI Z136.1, Safe Use of Lasers; IEC 60825-1, Safety of laser products. Check current editions and local applicability.